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HCF & LCM

Department of Training & Placement
Aditya University

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Introduction

Definition of LCM and HCF:

- **Lowest Common Multiple (LCM):** The **least or smallest** common multiple of any two or more given natural numbers are termed as LCM. For example, LCM of 10, 15, and 20 is 60.

- **Highest Common Factor (HCF):** The **largest or greatest factor** common to any two or more given natural numbers is termed as HCF of given numbers. Also known as GCD (Greatest Common Divisor). For example, HCF of 4, 6 and 8 is 2.

Key Formulae

HCF and LCM Formulas:

- LCM × HCF = Product of the Numbers
- LCM of Co-prime Numbers = Product Of The Numbers

$$\text{LCM of fractions} = \frac{LCM\ of\ numerators}{HCF\ of\ denominators}$$

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Questions

Ques.1 Find the LCM of 48 and 32.

a) 96

b) 16

c) 144

d) 156

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Questions

Ques.2 Find the greatest number that will divide 12, 36 and 54 exactly.

a) 2

b) 6

c) 1

d) 15

Questions

Ques.3 Find the LCM and HCF of 7/9, 63/8, 9/7.

a)63, 504

b)63,1/504

c)72,75

d)1/63,504

Questions

Ques.4 Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.

a)9

b)4

c)7

d)13

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Questions

Ques.5 Find the greatest five-digit number which is divisible by 7, 10, 15, 21 and 28.

a)99540

b)99960

c)99990

d)99988

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Questions

Ques.6 Six bells commence tolling together and toll at intervals of 2, 4, 6, 8 10 and 12 seconds respectively. In 30 minutes, how many times do they together.

a)4

b)10

c)12

d)16

Questions

Ques.7 The greatest number of four digits which is divisible by 15, 25, 40 and 75 is:

a) 9000

b) 9400

c) 9600

d) 9800

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Questions

Ques.8 Find the greatest number which when it divides 367, 135 and 259 will leave the same remainder in each case.

a) 4

b) 6

c) 8

d) 2

Questions

Ques.9 Three bells are heard at the intervals of 11, 22, 44 minutes respectively. If all bells toll together exactly at 10:16 A.M., they will again be heard together at _____.

a) 12:00 A.M

b) 11:00 P.M

c) 11:00 A.M

d) 11:40 A.M

Questions

Ques.10 Three number are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:

a) 80

b) 40

c) 120

d) 160

Questions

Ques.11 What should be the maximum capacity of the measure to draw 3456, 1926 and 198 litres from three different cans?

a) 6

b) 12

c) 18

d)24

Questions

Ques.12 The ratio of two numbers is 13:7 and their HCF is 15. What is the sum of two numbers?

a) 50
b) 66
c) 44
d) 300

Questions

QUES.13 Find the least number which when divided by 5, 7 and 11, leaves a remainder 6 in each case.

a) 431
b) 391
c) 378
d) 385

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Questions

QUES.14 Find the greatest number which when it divides 714, 435 and 615 will leave 3 as remainder in each case.

a)15
b)12
c)9
d)14

Questions

QUE.15 The ratio of two numbers is 3 : 4 and their H.C.F. is 4. Their L.C.M. is:

a) 12
b) 16
c) 48
d) 24

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Thank You